

Research Article

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Rough Fabry–Perot cavity: a vastly multi-scale numerical problem

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Abstract: A commercial Fabry–Perot laser diode is characterized by highly disproportionate dimensions, which poses a significant numerical challenge, even for state-of-the-art tools. This challenge is exacerbated when one of the cavity mirrors is roughened, as is the case when fabricating random laser diodes. Such a system involves length scales from several hundred micrometres (length) to a few nanometres (roughness) all of which are relevant when studying optical properties in the visible. While involving an extreme range of dimensions, these cavities cannot be treated through statistical approaches such as those used with self-similar fractal structures known to show well-studied properties. Here we deploy numerical methods to compute cavity modes and

show how random corrugations of the Fabry–Perot cavity wall affect statistical proper-ties of their spectral features. Our study constitutes a necessary first step in developing technologically essential devices for photonic computation and efficient speckle-free illumination.

Keywords: disorder photonics; numerical modelling; optical cavities; random lasers; mixed length scales; analytical models

1 Introduction

Diffusive elements in optics can be seen as detrimental features as they typically lead to increased attenuation, uncontrolled light propagation, and reduced coherence [1], [2]. However, it can be advantageous for the realization of non-conventional light emitters, like random lasers (RLs) [3], in which scattering elements provide optical feedback for lasing.

Random lasers have been firstly proposed as “non-resonant” [4] lasers in which one of the mirrors of a Fabry–Perot cavity (FPC) was substituted for a scattering surface. First observations of lasing emission have been obtained in dye-doped colloidal suspensions [5] and in semiconductor powders [6] *i.e.*, in structures where the feedback is spatially distributed via a random spatial distribution of the refractive index.

Distributed feedback being the most common architecture considered in experimental realizations, theoretical investigations have been oriented to this kind of devices. Several theories [7] have been proposed approaching the problem as an exceptional case that can be understood with assumptions specific to RLs, *e. g.*, open cavities with quasi-bound states [8], extended amplified modes [9], closed feedback loops [10] or Anderson localization [11].

We recently proposed a unified taxonomy of lasing cavities, to include RLs, based on the nature of the cavity and the form of feedback: ordered/disordered structures and distributed/non-distributed feedback. This allows to classify them in four categories: FPC lasers, distributed

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feedback lasers and RLs with distributed and non-distributed feedback (see Supplementary Material in ref. [12]). We inferred this vision from the observation of the typical spectral emission of RLs from structures consisting of the active material, enclosed between two scattering elements [13] or a scattering element and a mirror [14]. Other authors also reported RLs with spatially separated regions for gain and feedback, *e.g.*, in active fibres with rough cladding [15], dried dye droplets with cracks [16] and polymers with scattering surfaces [17]. Previously referred to as RLs with spatially localized feedback, we propose the nomenclature of non-distributed feedback RLs or more generally of diffusive cavity lasers.

This class of emitters shows specific emission properties, which arise from the high numbers of optical modes existing in the cavity: multiple randomly distributed frequency peaks or single peak emission (interpreted as highly coupled modes), low spatial coherence and high divergence angle of emission. Different applications take advantage of these properties, *e.g.* high efficiency lighting [18], super spectral resolution spectroscopy [19], signal processing [20] and coding [21], photonic hardware for artificial intelligence [22], counterfeiting [23], and biomarkers [24]. This highlights the importance of understanding and controlling disorder in order to, through engineering, make a better use of it [25]. However, a clear and solid theoretical treatment of this class of lasers is still missing.

Here, we offer a theoretical analysis and perform numerical simulations of the structure consisting of a plain mirror and a scattering surface, like the one based on semiconductor materials obtained in previous work [12].

2 Problem description

Semiconductor laser diodes rely on a carefully engineered structure designed to achieve efficient operation. The active

medium, typically a multiple quantum well, is sandwiched between two doped layers that enable carrier injection. These layers, having a wider electronic band-gap and lower refractive index, also serve as confining regions for both charge carriers and optical modes. The device is terminated by cleaved edges, forming an FPC that provides strong feedback for in-plane propagation.

Two common FPC configurations help define lasing properties. In electrode-controlled devices, a top electrode selectively activates the lasing region, creating an area where current density exceeds the necessary threshold (see Figure 1a right drawing). An alternative approach confines the active medium in a ridge waveguide, preventing lateral current spread while enhancing both optical and electronic confinement (see Figure 1a left drawing). The system considered in this study consists of a high aspect ratio 3D FPC, with active layers only a few nanometres thick compared to device width of tens or length of several hundred micrometres. This dimensional imbalance allows to model the structure effectively as two-dimensional.

Even under the simplifying assumption of a two-dimensional geometry, the large in-plane dimensions of commercial-scale FPCs (on the order of tens of thousands squared wavelengths) make numerical treatment highly challenging. Beyond pure scale considerations, a further challenge in laser cavity modelling is the impact of random surface roughness, since accurately describing the surface corrugations significantly increases the degrees of freedom of the problem.

To provide context, finite-element method eigensolvers typically depend on sparse-matrix diagonalization algorithms, such as the Arnoldi method implemented in ARPACK [26]. When applied to large-scale optical cavities with few relevant modes, these algorithms exhibit quadratic scaling in both time and memory with respect to the number of degrees of freedom [27], [28]. Thus, it becomes apparent

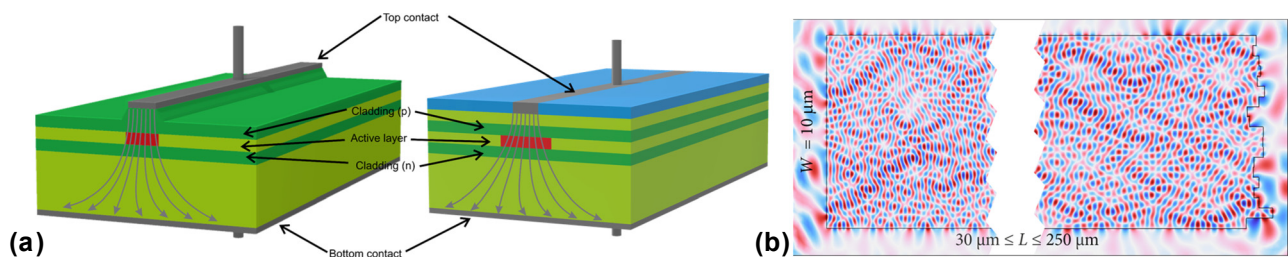


Figure 1: Typical FPC laser design and sample rough cavity mode. (a) General design of a semiconductor laser diode. Depending on the width and type of lateral confinement several morphologies can be conceived. The ridge waveguide structure (left) makes use of physical lateral confinement by including the active region in or near an etched ridge that directs the injected current (only or mostly depending on the height of the ridge) through a limited width of the recombination layer (left). In the right-hand side structure, the width of the active region is determined by the spread of the injected current. (b) Geometry of the FP cavity and simulation domain including detail of the rough FP edge composed of N_{sec} sections, each being assigned a random length uniformly fluctuating around a mean value. The wavefunction is color coded as usual: red positive blue negative, white zero.

how precisely modelling laser cavities with lengths of tens or hundreds of micrometres, with corrugations in the end mirrors featuring correlation lengths of hundreds of nanometres and depths of tens of nanometres, requires handling very different length scales with high accuracy. In this work, we employ the finite-element method solvers implemented in COMSOL® Multiphysics to address these numerical challenges and compute the eigenmodes of the system. Related problems although with disorder present in a distributed fashion have attracted attention [29].

We emphasize that in this study we focus on the passive (i.e., without gain) cavity modes of rough FPCs, which can offer valuable insight into behavior in the lasing regime.

It is worth mentioning at this point that in open 1D disordered systems with weak disorder two types of the eigenstates can be found: ordinary *quasi normal modes* which are associated with a transmission resonance, and *hidden* quasi normal modes which do not exhibit peaks in transmission. This distinction however, is absent in periodic systems and resonators where there is a one-to-one correspondence between them [30].

In these disordered cavities, optical modes initially experience gain compensation of roughness-induced scattering, followed by gain narrowing as they evolve toward the lasing state [31]. These lasing modes are defined through nonlinear interactions with the gain medium, and their direct numerical study remains a highly demanding task. Nevertheless, although lasing states do not strictly coincide with passive cavity solutions, they can be understood as emerging from them. Thus, analyzing the passive eigenmode spectrum offers a reasonable estimate of lasing frequency distributions. Furthermore, since these cavities exhibit inherent randomness, a statistical framework that describes their spectral properties and disorder-induced effects will also be valuable to describe the properties of lasing modes.

3 Numerical inspection

To study the resonant modes and frequency characteristics of passive FPCs, with a rough mirror, we perform an eigenfrequency analysis using COMSOL Multiphysics. The FPCs considered here (shown in Figure 1a) are dielectric slabs with refractive index $n_s = 3.86$ embedded in air. The slab has a fixed width, $W = 10 \mu\text{m}$, and we studied cases with a length, L , ranging from $30 \mu\text{m}$ to $250 \mu\text{m}$. To introduce surface roughness in a controlled way while holding computational cost in check, we applied a random corrugation to the right-hand side boundary of the dielectric slab. This corrugation is in the form of a simple crenelated profile

consisting of N_{sec} sections of different length (see Figure 1b). All sections have the same width but vary in length leading to steps of varying depth in the edge. These depths are drawn from a uniform random distribution in the interval $[0, h]$, where h defines the maximum fluctuation amplitude considered in a realisation. The uniform distribution used here was chosen for its simplicity and general applicability, without relying on assumptions about the microscopic origin of the surface roughness. For the eigenfrequency analysis, we specified an eigenvalue search range centred around 500 THz and solved for the 50 closest eigenfrequencies, allowing us to capture a comprehensive set of resonant modes near this frequency. This setup provides insights into the cavity's mode structure and frequency response under different levels of boundary roughness. All results discussed in this work pertain to transverse electric (TE) polarization. Similar results are obtained for transverse magnetic (TM) polarization as shown in the Supplementary Material.

3.1 Average spectral distance

The average spectral separation, $\langle s \rangle$, a dimensionless spacing between adjacent eigenfrequencies, is calculated as a normalised average of the mode spacings $\Delta f_n = f_{n+1} - f_n$:

$$\langle s \rangle = \frac{1}{N} \sum_{n=1}^{N-1} \frac{f_{n+1} - f_n}{f_n} \quad (1)$$

where N is the number of eigenfrequencies considered, f_n , calculated in a given frequency interval.

Figure 2a illustrates the average spectral separation, $\langle s \rangle$, as a function of roughness depth, h , of the corrugation and for different number of the sections, N_{sect} , for a cavity of length $L = 30 \mu\text{m}$. Across the entire range of N_{sect} and h , $\langle s \rangle$ fluctuates around the baseline given by the free spectral range (FSR) of the regular FPC shown as a horizontal solid line. This indicates that disorder introduces variability in spectral spacing but generally does not lead to substantial modifications of the average spectral separation of the regular FPC. In Figure 2b, we set $N_{\text{sect}} = 40$ and show $\langle s \rangle$ as a function of h for cavities of lengths 30, 60, 150 and $250 \mu\text{m}$, shown in purple, green, blue, and red, respectively. The normalized averaged spectral separations again fluctuate around the FSR of the corresponding regular FPCs. These reference FSR values are indicated by horizontal lines. Since h does not seem to have a strong impact, we take the average for all h and plot $\langle s \rangle$ as a function of L in Figure 2c. One can see that $\langle s \rangle$ decreases as L increases. This reflects the expected inverse proportionality between the FSR of an FPC and its length, $f_{\text{FSR}} \propto 1/L$. In SI we include a simple argument to account for the $1/L$ dependence whose prediction is marked in Figure 2c as a dashed red line. Under that

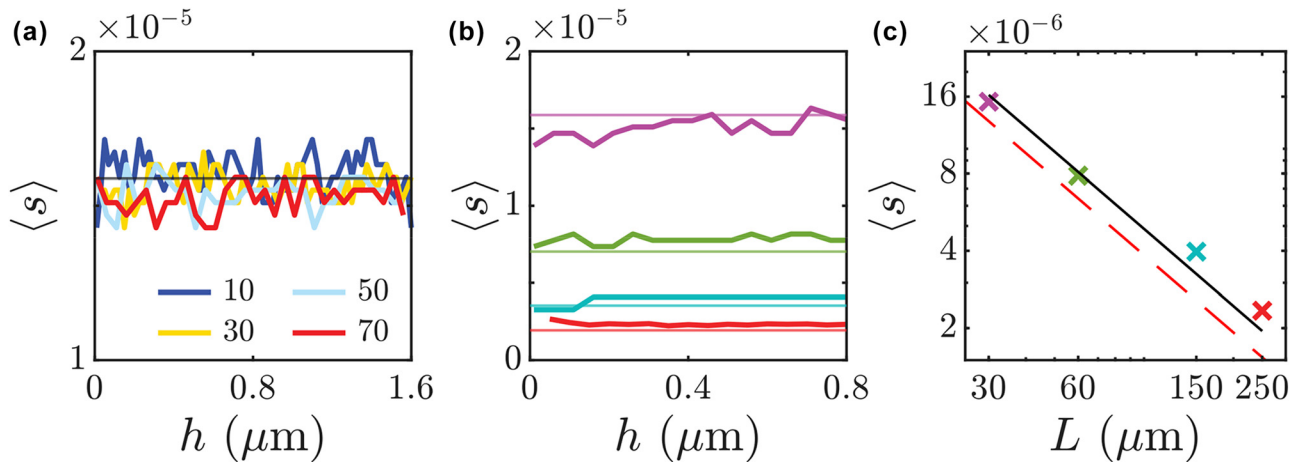


Figure 2: Spectral separation dependence on cavity features: numerical results. (a) Dimensionless average spectral separation, $\langle s \rangle$, as a function of roughness depth, h , for different number of the sections N_{sect} for an FPC with a randomly corrugated wall. (b) Same as a function of roughness depth h for FPCs with various lengths L . Different coloured lines correspond to cavities with $L = 30$ (purple), 60 (green), 150 (blue) and 250 (red) μm , respectively. Horizontal lines correspond to $\langle s \rangle$ in the pristine case for the different cavity lengths. (c) Mean (in h) average spectral distance as a function of FPC length in log-log scale. The black solid line represents the fit of the numerical data with analytical expression $\langle s \rangle = 4.87 \times 10^{-10}/L$ and the red dashed line a simple model based on mode count in reciprocal space (see SI). $N_{\text{sect}} = 40$ for panels (b) and (c).

assumption, we fit the numerical data for mean (in h) average spectral separation to an analytical expression, finding $\langle s \rangle = 4.87 \times 10^{-10}/L$. The analytical results correspond to the black solid line. They show good agreement with the numerical data, represented by crosses, over the considered range.

3.2 Average spectral broadening

The dimensionless mode spectral broadening – indicating the spread of resonance linewidths – can be calculated as the average of the modes' inverse quality factors:

$$\langle Q^{-1} \rangle = \frac{1}{N} \sum_{n=1}^N \frac{1}{Q_n} \quad (2)$$

where $Q_n = f_n/\delta f_n$ with δf_n the mode width. In Figure 3a we show that $\langle Q^{-1} \rangle$ initially sharply increases with h for any number of sections (cavity of length $L = 30 \mu\text{m}$). Beyond a certain threshold, around $h \approx 0.5 \mu\text{m}$, $\langle Q^{-1} \rangle$ values begin to stabilize, particularly for the larger numbers of sections. This plateau suggests that, after an initial high sensitivity to roughness, the spectral broadening reaches a level where further increases in h does not lead to a proportional

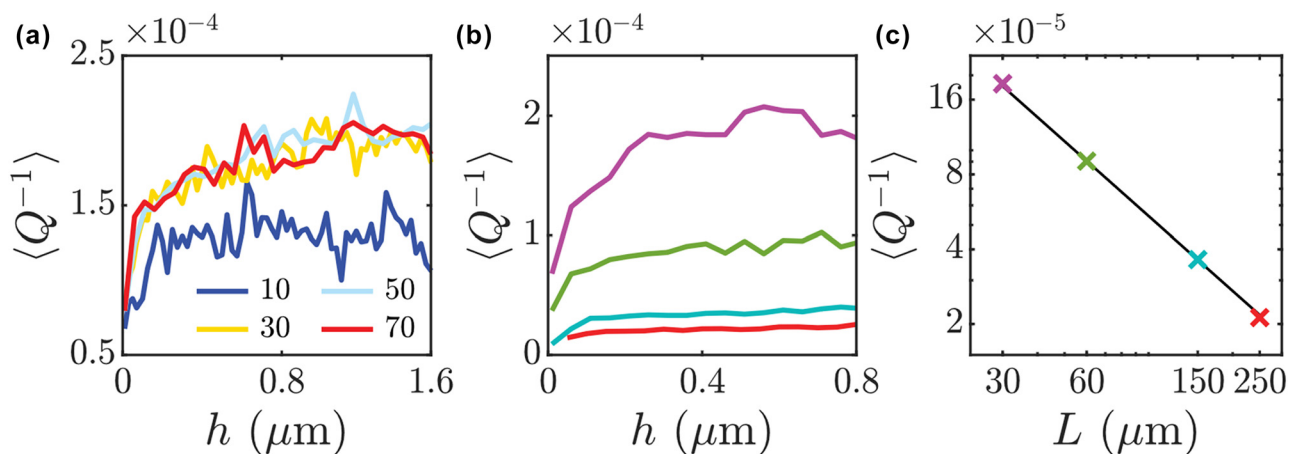


Figure 3: Spectral broadening dependence on cavity features: numerical results. (a) Dimensionless average spectral broadening $\langle Q^{-1} \rangle$ as a function of roughness depth, h , for different number of the sections N_{sect} . Cavity length is 30 μm . (b) Dimensionless average spectral broadening as a function of roughness depth for different FPC lengths, L : 30 μm (purple), 60 μm (green), 150 μm (blue) and 250 μm (red), respectively. (c) $\langle Q^{-1} \rangle$ averaged over roughness depth (h), as a function of FPC length, shown on a log-log scale. $N_{\text{sect}} = 40$ for panels (b) and (c). The black solid line represents the fit of the numerical data analytical expression $\langle Q^{-1} \rangle = 5.43 \times 10^{-3}/L$.

increase in $\langle Q^{-1} \rangle$. In Figure 3b, the number of sections is fixed to $N_{\text{sect}} = 40$, and we show $\langle Q^{-1} \rangle$ as a function of h for cavity lengths of 30, 60, 150 and 250 μm , given respectively by the purple, green, blue, and red lines. All traces follow the same structure, with a sharp initial rise and then a plateau onto a stable value, indicating that broadening saturates for all cavity sizes. In Figure 3c we plot the estimated asymptotic broadening value for the different cavities in Figure 3b. This figure illustrates that $\langle Q^{-1} \rangle$ decreases as L increases, indicating that quality factor reduction is related to the quality factor of the pristine cavity. Figure 3c further reinforces the inverse relationship between L and $\langle Q^{-1} \rangle$ displaying a clear downward trend as L increases. This inverse mean length dependence suggests that regardless of roughness levels longer cavities retain higher quality modes as their pristine counterparts.

It is worth noting that the results presented in Figures 2 and 3 correspond to a single representative realization of the random wall roughness. This is sufficient to capture the characteristic trends, as additional calculations (see Figures S7 and S8 in the Supplementary Information) demonstrate that different realizations produce very similar spectral behaviour.

4 Analytical model

In the previous section, we demonstrated that roughness leads to a reduction of the quality factor of resonances that saturates to a limiting non-zero value, while retaining the free spectral range of the unmodified cavities, *i.e.* without the introduction or removal of modes. In order to shed light onto this behaviour, we now introduce a simple analytical approach based on a 1D cavity, with one pristine mirror, with reflection coefficient r , and one rough mirror. The core assumption behind the model is that upon incidence on the rough mirror, waves will be reflected in a superposition of different phase-offsets, ϕ , where the weight of the different phase components is *normally* distributed around zero phase offset with σ_ϕ standard deviation in a way similar to ref. [13]. Here we are assuming that the magnitude of the amplitude coefficient for reflection, r , is merely determined by the index contrast and the phase change upon reflection, ϕ , obeys a Gaussian distribution of variance σ_ϕ^2 . That is to say, the probability with which an incident wavefront is reflected with a phase ϕ is given by $P(\phi) = \frac{1}{\sqrt{2\pi}\sigma_\phi} e^{-\frac{\phi^2}{2\sigma_\phi^2}}$. Under this assumption computing the expected value of $e^{-i\phi}$ is equivalent to calculating the Fourier transform of a Gaussian which is a Gaussian. So, it can be shown (see Supplementary Material) that the expected value for the

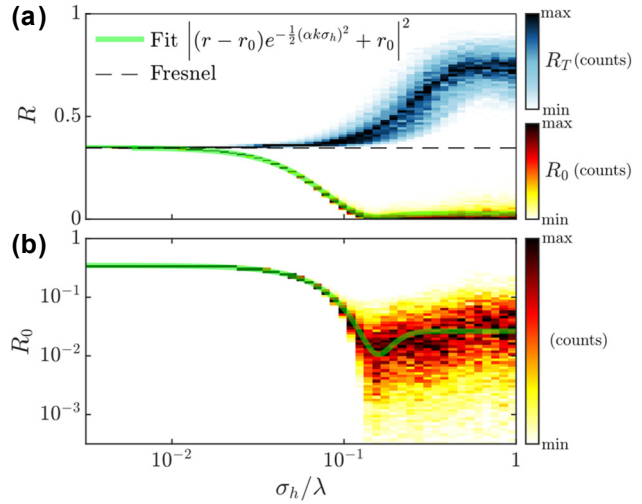


Figure 4: Statistical study of reflectance from a rough flat surface. (a) 2d histograms of total (blue-black) and specular (red-black) reflectance from a rough surface. (1k realizations per roughness level) together with a fitting to analytical expression. For each roughness level, the most frequent value is shown in black and the least in white for both magnitudes. (b) Histogram of specular reflectance now in log-scale, revealing the saturation of the value of the specular reflectivity for large roughness.

amplitude reflectance coefficient is $r' = re^{-\frac{1}{2}\sigma_\phi^2}$, which shows that this phase superposition, through destructive interference of the reflected fields, effectively acts as a loss mechanism for the FPC. We further assume that the standard deviation of the phase distribution can be related to the surface's RMS roughness, σ_h , by, $\sigma_\phi = \alpha k \sigma_h$ where $k = 2\pi n_s/\lambda$ is the wavevector in the cavity and α is an a dimensional factor related to the sensitivity of the particular mode to the roughness.

To test this model, we perform numerical simulations in COMSOL measuring the power reflected from a rough surface. In sight of Figures 2c and 3c we believe numerical estimates can be safely extrapolated to large L so, throughout this section, in order to study the statistical properties of these cavities, we will focus on FPCs of smaller size. This alleviates the massive computational effort required to model roughness in a more realistic manner, closer to experimental realizations. In particular, roughness is implemented by superposition of random waves with correlation lengths of 100 nm and a controlled root mean square (RMS) displacement. Figure 4 presents a statistical analysis of the reflected power from a rough surface, based on 1,000 realizations per roughness level. In the panels, we distinguish between specular (R_0), and total integrated reflectance (R_T), obtained for a plane wave of $\lambda = 600$ nm. The wave is normally incident from a medium with $n_s = 3.86$ onto the rough surface at the boundary with vacuum. The colourmaps render

histograms of the numerically-obtained reflectivity values, ranging from white (least frequency, minimum number of counts) to black (highest frequency, maximum number of counts) for both quantities. These panels enable us to infer the probability distribution for the underlying stochastic variable and determine the most likely values for R_0 and R_T as a function of the mirror roughness. In Figure 4a, and in the limit of low roughness, the reflectance is well described by the Fresnel reflection coefficient (horizontal dashed line). However, as roughness is increased beyond $10^{-2}\lambda$, specular reflectance values drop off significantly, reaching vanishing values for roughness levels of $10^{-1}\lambda$. In Figure 4b we plot the specular reflectance in logarithmic scale to reveal that it plateaus to a limiting value at high roughness levels. To capture this behaviour, we include a phenomenological remnant reflectance, r_0 , in our model, which will dominate the rough-cavity limit. Therefore, we write the reflectance from the rough surface as

$$r' = \left| (r - r_0) e^{-\frac{1}{2}(\alpha k \sigma_h)^2} + r_0 \right|^2 \quad (3)$$

in qualitative agreement with the seminal works by Bennett [32] and Davies [33] on surface-roughness-induced variations in reflectance, whose formulations remain widely used in contemporary analyses of optical systems.

In Figure 4 we show the fitting (using 2 fitting parameters, r_0 and α) of the simulated reflected values to this distribution as a green line in both panels, demonstrating that this phenomenological approach correctly captures the degradation of the feedback mechanism that will ultimately lead to the general broadening of cavity resonances showcased in the previous section. The best fit in this case corresponds to $r_0 = 0.16$ and $\alpha = 1.87$, which predicts a residual reflectance of the rough mirror of $|r_0|^2 = 2 \times 10^{-2}$. Furthermore, note that for the simple case of normal incidence, the difference of optical path for a wave impinging on a boundary offset by σ_h would be of approximately $2\sigma_h$ and, therefore, from simple geometric arguments, one could expect a value of the sensitivity parameter, α close to 2, as we find in the fit.

Once the reflection coefficient from the single scattering event is established, as shown in the Supplementary Material, we use it to develop an analytical solution for the fields scattered by a 1D cavity in which one mirror is pristine, with coefficient r , and the other mirror is roughened, with reflection coefficient r' . The treatment can be readily extended to more realistic Fabry–Perot cavities, where the front and back mirrors may have differing reflectivities, as is commonly the case in conventional diode lasers. Based on this analytical model, we can determine the resonant

modes of the cavity, and by performing a small frequency expansion around these resonances, one may find the usual associated Lorentzian lineshapes, and also an expression for the quality factor of such resonances. See Supplementary Material for details. We can write the expression for the quality factor of the n -th cavity mode in the following compact form:

$$Q_n = \frac{Q_n^{(p)}}{\sqrt{1 + \frac{(1-\eta)}{2\eta} [F_n^{(+)} - F_n^{(-)}\eta]}} \quad (4)$$

where $Q_n^{(p)}$ is the quality factor in the pristine limit and $F^\pm$ are defined as $F_n^{(\pm)} \equiv (1 + A_n^2 \pm A_n \sqrt{2 + A_n^2})/A_n^2$, where $A_n = (\sqrt{2n\pi})/Q_n^{(p)}$ is related to the mode's order, n , and unperturbed quality factor. Finally, the variable in Eq. (4) that introduces the effect of roughness is the η factor, defined as

$$\eta = e^{-\frac{\sigma_\phi^2}{4}} \left[\cosh\left(\frac{\sigma_\phi^2}{4}\right) - \sinh\left(\frac{\sigma_\phi^2}{4}\right)(1 - 2\beta) \right] \quad (5)$$

which depends on the roughness level through $\sigma_\phi = \alpha k \sigma_h$, as discussed above. Furthermore, note that this factor also contains the effect of saturation of reflectance losses through $\beta \equiv r_0/r$. This term leads to a saturating behaviour in the quality factor too, as observed in Figure 4 which, in the limit of high roughness, tends to a saturation value obtained from Eq. (4) by replacing η by β . This directly reproduces the numerically observed fact that the saturated Q factor for the rough cavities is proportional to cavity length (since the quality factor in the pristine case follows this proportionality).

Note that in this model, although the quality factor is modified, as compared to the pristine case, the resonance frequency of the different modes is not affected. This is a direct consequence of the 1D, single mode nature of the model, and constitutes a significant simplification. More realistic characterization would involve considering how roughness mixes different modes of the 3D cavity, which would lead to mode hybridization and frequency shifts [34]. Nevertheless, as we will show in the next section, the exact resonance frequency of a given mode is not particularly relevant for the statistical analysis. As shown in the Supplementary Material, a first-order perturbative treatment predicts that the expectation values of the resonance frequencies remain unchanged to leading order in roughness. For this reason, our simplified model adequately captures the degradation of resonances resulting from complex mode mixing.

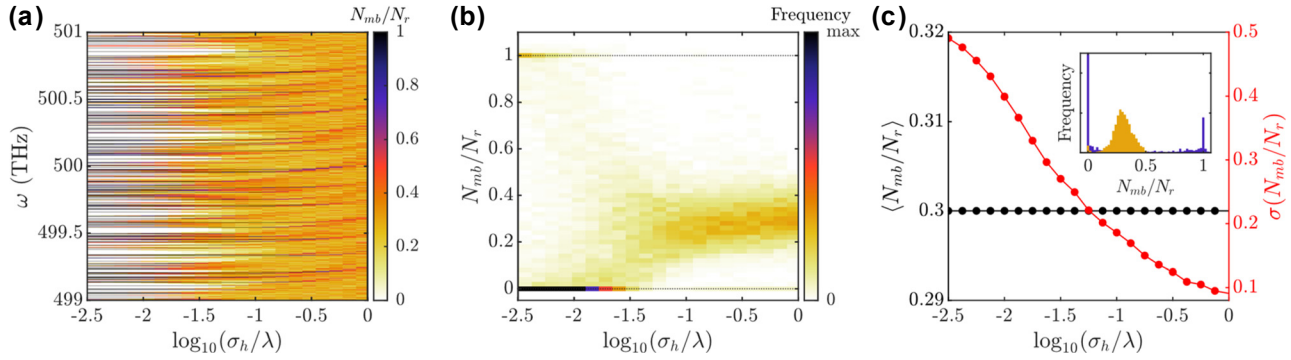


Figure 5: Statistical analysis of the spectral properties of rough FP cavities. Panels a–c correspond to cavities of size $10 \times 30 \mu\text{m}$. (a) Two-dimensional map presenting the number of modes in the whole realizations set per frequency interval, normalized to number of cavity realizations as a function of frequency and roughness. (b) Frequency or number of occurrences in which the bins of panel a contain N_{mb}/N_r modes as a function of roughness. Colour scale saturates to better showcase a feature at $N_{mb}/N_r \approx 0.3$. (c) Ensemble-averaged and normalised number of modes per frequency interval (bin), and standard deviation. The inset shows line cuts of the distribution of panel b for roughness values $\log(\sigma_h/\lambda) = -2.5$ (purple) and 0 (yellow).

5 Statistical properties of rough FPC

In what follows, we present a statistical study of 2D FPCs of fixed size, $10 \mu\text{m}$ by $30 \mu\text{m}$, in which roughness is modelled through a random corrugation of the same characteristics as used for Figure 4. To obtain the data we generate $N_r = 50$ random realizations of cavities for each roughness level and record the eigenfrequencies in a 2 THz interval around a central frequency of 500 THz. We will follow the same structure as Section 2: first looking at the properties of the spectral distribution and then proceeding to analyse the effect that roughness has on the quality factors of the modes.

In Figure 5a we show a 2D histogram where we divide the 2 THz frequency window into one-thousand bins of 2 GHz width (which is narrower than the pristine cavity's FSR of approximately 7 GHz implying that roughly three out of four bins must be empty in a typical realization). For each frequency bin and each roughness level, we plot the number of modes in the bin, N_{mb} , as a function of roughness averaged over realizations. Thus N_{mb}/N_r allows us to infer the probability that FPC contains a mode within a 7 GHz window around the central frequency of the bin. Note that here we are using that there is no FPCs presenting more than 1 mode within a single frequency window. For low roughness levels, the mode distribution is sharply structured, always following the pristine case. As roughness is increased, these eigenfrequencies undergo slight shifts in spectral position and broadening, leading to a smearing out of the initially spiky line-shape. Further increasing the roughness leads to ever smoother spectra, which tends to a uniform distribution in the high roughness limit. The modal

spectrum variation with increasing roughness can be seen in the colour map in Figure 5b. It shows the frequency with which bins contain a certain number of modes N_{mb}/N_r , as a function of the mirror roughness normalized to the number of cavity realizations. For very low levels of roughness, the underlying probability distribution is bimodal, with a dominant peak at $N_{mb}/N_r = 0$ and a small one at $N_{mb}/N_r = 1$. That is, in this regime, the vast majority of the frequency intervals are empty in all realisations (making the frequency peak at $N_{mb}/N_r = 0$); only a few containing exactly 1 mode (giving rise to a secondary peak at $N_{mb}/N_r = 1$), the rest of occupancies showing a small frequency.

As roughness increases, more frequency windows in the whole realisation set contain modes, as shown by the smearing out of the initially sharp feature at $N_{mb}/N_r = 0$. Now, intermediate values of N_{mb}/N_r appear with significant frequency, thereby reducing the incidence of zero or unity ratios. For high roughness, a pattern resembling a normal distribution emerges in the frequency map, with a mean value of $\langle N_{mb}/N_r \rangle \approx 0.3$ and a rather small standard deviation, $\sigma(N_{mb}/N_r)$, of the order of 0.1. Thus, in the rough-cavity limit, all frequency bins contain roughly the same number of modes, which agrees with a uniform frequency distribution at high σ_h in Figure 5a.

Figure 5c presents the statistical analysis of the values of N_{mb}/N_r , across different roughness levels, rendering both its mean value and standard deviation. The mean value, $\langle N_{mb}/N_r \rangle$, remains constant despite variations in roughness, indicating that the total number of modes within the fixed frequency window (499–501 THz) is invariant, and thus showing that the FSR of the cavity is unaltered by roughness. In contrast, the standard deviation decreases as roughness increases. This should not be surprising considering that for the lowest roughness values the distribution is bimodal

peaking a zero and one and maximizes the standard deviation $\sigma(N_{mb}/N_r)$. These results are consistent with a perturbative treatment of the effect of surface roughness on the resonance frequencies (see Section 5 of the Supplementary Materials), which shows that for normally distributed corrugations the resonance frequencies remain, on average, unaltered, leaving the FSR essentially constant. In contrast, systematic biasing of the roughness spectrum can lead to net frequency shifts for specific modes. This reasoning also extends to more realistic cavities with reduced lateral confinement stemming from lower refractive-index contrast (see Section 3 of the Supplementary Materials).

At low roughness many frequency bins are empty because mode width is narrower than bin span, but as roughness grows the discrete mode spectra transitions into the rough cavity limit, where modes spread and frequency bins present a normally distributed occupancy. This trend highlights a transition in the behavior of the FPC spectral features, where increased disorder leads to statistical smoothing of spectral characteristics. Additionally, the inset of Figure 5c presents line cuts of the N_{mb}/N_r distribution extracted from Figure 5b, illustrating the two limiting cases of $\sigma_h/\lambda = 10^{-2.5}$ and $\sigma_h/\lambda = 1$. These visualizations further emphasize the statistical evolution of mode distributions under increasing roughness.

We now shift our attention to the behaviour of the quality factor of the rough cavity modes. In Figure 6a we show the histogram of the number of modes with a given quality factor for each roughness level. In this case, we divide the Q -factor axis into 150 bins, and display in colour the number of modes per bin, N_{mb} , normalized to the number of rough cavity realizations at each roughness level, $N_r = 50$. In this map one readily sees that, for low roughness levels, two sets of quality factor values appear frequently, which we

associate to two groups of FPC modes: those with a very narrow distribution of low Q values, and those with broader range of – two orders of magnitude – higher quality factors.

For both groups of modes, it is *ostensible* that Q is almost unmodified for σ_h below $\sim 0.01\lambda$. Beyond that, the modes with highest quality factors, and therefore most sensitive to additional loss mechanisms, experience a clear drop in Q (shortening of their lifetime), which ultimately saturates to a fixed value for $\sigma_h \approx \lambda$. Interestingly, the group of modes with $Q < 10^3$ is completely unfazed by the introduction of roughness. This can be understood by considering roughness as another loss mechanism: in the regime in which the mode broadening is governed by other phenomenon, *e.g.*, scattering or material absorption, the extra losses stemming from roughness may be negligible. Furthermore, one can also observe how some modes with initially low- Q experience a slight increase in their quality factor. This we link to their roughness-induced hybridization with higher- Q modes, with which they interact to form a low- Q tail in the distribution at high roughness levels. As mentioned before, this cannot be modelled by our simple analytical treatment, since we are assuming fully independent modes. Nevertheless, this effect remains minor. In Figure 6b we show the ensemble-averaged quality factor (consider all groups of modes introduced above), together with the fit to the analytical expression in Eq. (4), demonstrating an excellent agreement. Furthermore, in Figure 6c we also fit the analytical expression to the average quality factors obtained from cavities larger than $L = 150 \mu\text{m}$ (as those considered in Figure 2), also demonstrating a good agreement. These panels show that our simple analytical model can provide valuable insight into the modification of the FPC modes as roughness is introduced.

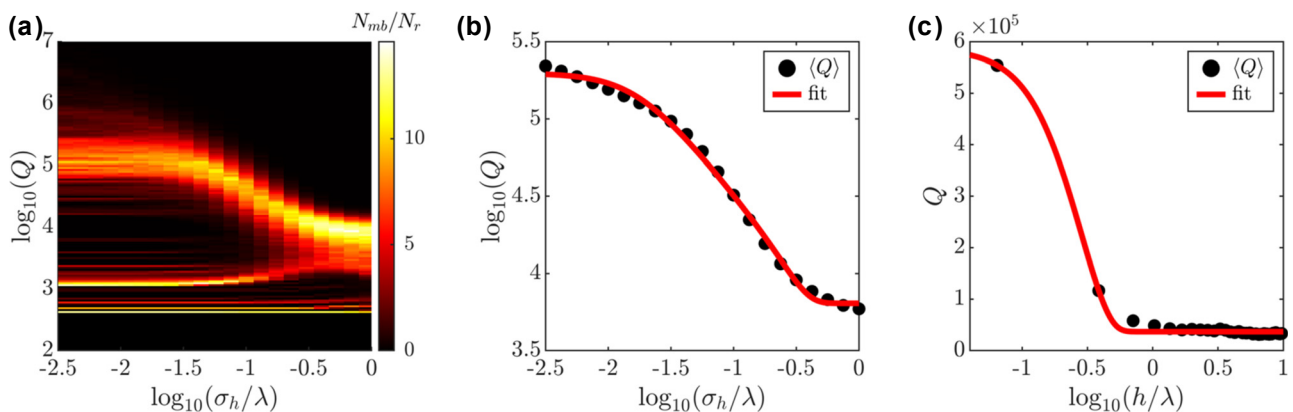


Figure 6: Statistical analysis of rough FP cavities. (a) Two-dimensional histogram showing the number of modes for a given roughness and quality factor. (b) Mean quality factor as a function of roughness and fitting to the analytical model. Panels (a–b) correspond to cavities $10 \times 30 \mu\text{m}$. (c) Same as panel (b) for data corresponding to larger cavity of size $10 \times 150 \mu\text{m}$.

6 Conclusions

In summary, we have carried out numerical simulations of vastly multiscale FPCs involving dimensions from sub-millimetre (length) to nanometre (depth) scales to study the effect of random roughness with correlation lengths of the order of sub-micrometre. The simulations show that spectral mode distributions, allowing for the fluctuations inherent to successive realisations, follow approximately the perfect FPC free spectral ranges without much dependence on roughness depth. These also yield, as expected, a compression of the FSR approximately following a $1/L$ trend as the average length of the cavity grows. The quality factor of the modes, after a fast start, decrease slowly as the roughness of the output mirror increases.

Most of these features are captured by an analytical model that considers the reflection on the rough FPC mirror as a source of normally distributed phase that, through interference, permits treating the roughness as the origin of a reduced reflectance. Assuming that the phase distribution is associated to the roughness depth, this model allows to analytically compute effective reflectance and quality factors, all in very good agreement with numerical results. Roughness related loss of quality saturates leading to a residual value for very high roughness.

We expect that this kind of studies will allow a deeper understanding of the impact of fabrication on optical performance of devices [14] and guide future work.

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Data availability: The data that support the plots within this paper are available from the corresponding authors upon

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